

IZMAIL, Ukraine

WMO: 338890

Lat: 45.3703N

Lon: 28.8508E

Elev: 98

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	13.9	2.9	6.4	11.0	7.5	8.1	14.9	30.2	27.3	26.7	27.6	6.6	0	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	91.1	70.3	88.5	69.5	85.7	68.5	72.7	85.2	71.4	83.8	70.1	82.2	8.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	106.8	77.9	67.4	101.0	76.7	65.9	95.9	75.7	36.3	85.1	35.2	84.1	34.1	82.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.8	19.7	17.5	DB	3.9	96.0	4.3	4.0	0.8	98.9	-1.7	101.2	-4.1	103.5	-7.3	106.4	
			WB	3.1	75.5	4.3	1.9	0.0	76.9	-2.5	78.0	-5.0	79.1	-8.1	80.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	30.7	35.2	42.4	52.6	62.4	69.5	74.4	73.1	63.6	53.3	44.7	32.2
	DBStd	16.83	8.52	9.32	7.56	6.71	5.79	4.73	4.50	5.50	6.33	7.72	8.48	9.04
	HDD50	2131	599	416	257	49	2	0	0	0	1	53	199	555
	HDD65	5208	1064	834	702	375	117	13	1	4	99	372	609	1018
	CDD50	3201	0	2	20	127	388	586	757	717	409	154	40	1
	CDD65	804	0	0	0	3	37	149	293	256	58	8	0	0
	CDH74	7344	0	0	2	38	365	1246	2832	2431	385	45	0	0
	CDH80	2469	0	0	0	5	54	313	1086	944	65	2	0	0
(14) Wind	WSAvg	7.3	7.0	7.6	8.9	8.3	7.3	7.5	7.1	6.7	7.0	6.7	6.8	7.1
(15) Precipitation	PrecAvg	18.8	1.2	1.5	1.3	1.4	1.8	2.2	2.0	1.5	1.8	1.1	1.5	1.6
	PrecMax	30.6	6.4	4.7	3.7	3.4	5.0	5.7	7.0	4.3	6.1	3.5	4.5	6.3
	PrecMin	11.7	0.0	0.1	0.0	0.2	0.2	0.2	0.0	0.2	0.1	0.1	0.1	0.1
	PrecStd	3.9	1.1	1.1	1.1	0.7	1.2	1.3	1.5	1.1	1.8	0.8	1.1	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	61.6	70.2	78.9	84.9	90.2	94.6	95.2	85.5	78.9	68.6	55.0
		MCWB	46.4	51.2	54.5	62.5	65.8	68.9	71.7	71.9	67.8	65.0	57.7	49.6
	2%	DB	48.0	55.8	63.1	72.8	81.0	86.5	91.3	91.5	81.6	73.5	63.2	50.7
		MCWB	43.1	48.0	51.0	58.6	64.4	68.5	70.7	70.0	65.4	62.8	55.2	46.5
	5%	DB	44.7	51.5	58.4	68.0	78.1	83.8	88.9	88.6	78.6	68.7	59.3	47.0
		MCWB	41.6	45.0	48.6	56.0	63.0	67.6	70.1	69.2	64.8	60.0	53.2	43.4
	10%	DB	41.6	47.2	54.2	64.6	75.0	81.0	86.1	85.5	75.2	65.0	56.2	43.7
		MCWB	39.1	42.6	46.2	53.8	61.7	66.1	69.1	68.1	63.1	57.8	51.9	40.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	52.2	55.1	63.8	68.9	72.2	74.8	75.8	70.6	67.2	59.0	51.3
		MCDB	52.9	60.2	67.9	77.3	81.4	83.4	88.8	90.1	79.4	75.1	65.8	53.5
	2%	WB	43.8	48.2	51.9	59.6	65.9	70.5	72.9	72.8	68.4	64.1	56.4	46.5
		MCDB	47.0	55.2	61.7	70.6	77.1	82.2	86.6	84.2	77.5	71.2	61.8	49.9
	5%	WB	41.5	45.5	49.3	57.1	64.2	69.1	71.6	71.3	66.4	61.4	54.3	43.6
		MCDB	44.3	50.6	57.3	65.5	74.8	81.0	85.0	83.3	74.4	67.8	58.0	46.5
	10%	WB	39.2	43.3	47.0	55.2	62.7	67.5	70.4	69.7	64.6	58.7	52.6	41.0
		MCDB	41.7	47.0	52.9	63.0	73.0	78.5	83.3	81.3	72.8	64.1	55.9	43.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	13.1	14.3	17.1	19.6	19.2	20.0	19.7	17.8	15.8	11.9	10.8
		MCDBR	14.2	19.4	21.7	23.4	24.3	23.3	24.0	24.3	22.2	20.8	17.9	14.9
		MCWBR	10.4	12.4	12.0	12.1	11.2	10.1	9.1	9.5	10.0	11.0	10.8	11.0
	5% WB	MCDBR	13.3	18.7	20.4	20.6	22.2	20.5	21.5	20.9	19.7	19.4	15.0	13.6
		MCWBR	10.0	12.4	12.1	11.5	11.1	9.5	9.1	9.1	10.0	10.8	9.9	10.5
(40) Clear-Sky Solar Irradiance	taub		0.341	0.371	0.403	0.441	0.418	0.417	0.437	0.446	0.412	0.386	0.363	0.341
	taud		2.298	2.236	2.196	2.118	2.266	2.296	2.253	2.218	2.279	2.328	2.334	2.322
	Ebn at Noon		236	251	259	259	271	271	263	255	254	243	226	222
	Edh at Noon		26	33	40	47	42	41	42	42	36	30	25	23
(44) All-Sky Solar Radiation	RadAvg		383	636	1042	1515	1919	2038	2032	1824	1321	830	437	325
	RadStd		64	89	97	137	123	118	111	95	109	74	64	53

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (33 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	+126	

Nomenclature: See separate page